

DANIEL ROCCA

Lead AI & Full Stack Software Engineer

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SUMMARY

Senior Full-Stack & AI Engineer with **15+ years of software engineering experience** building production-grade applications, cloud platforms, and intelligent systems. Deep expertise in **Python, FastAPI, Agentic AI, LLM orchestration, RAG, APIs, distributed systems, and cloud-native architecture**.

Hands-on experience delivering enterprise AI workflows with **LangGraph, OpenAI, Gemini, and Claude**, including multi-agent systems, human-in-the-loop learning, structured processing, evaluation, guardrails, and production observability.

Experienced in taking AI solutions from **architecture and prototyping through deployment and production support**, combining strong Python/backend engineering with full-stack, data, and cloud expertise to build scalable AI applications, retrieval systems, and enterprise automation.

EXPERIENCE

11/2023 – 12/2025

Parsippany, NJ

- Lead AI & Full-Stack Software Engineer

Origami Studios

- Led architecture and full-stack delivery across **MyTutor's UK and U.S. platforms**, emphasizing scalable backend services, APIs, automation workflows, and cloud-native integration across four product streams.
- Helped evolve the platform toward **AWS-based microservices, serverless components, and event-driven architecture**, improving scalability, data-flow reliability, integration flexibility, and release readiness.
- Delivered production capabilities across **backend services, scheduling automation, authentication, payments, analytics, virtual classrooms, administration, and third-party APIs** for complex multi-role workflows.
- Contributed to a platform serving **250K+ learners, 1,500+ schools, and 4M+ lessons**, working across architecture, implementation, testing, deployment, and production support.

01/2020 – 09/2023

Hoboken, NJ

- Full-Stack AI Software Engineer

Walmart Global Tech

- Led production delivery of a **Python-based Agentic AI financial reconciliation platform**, automating transaction matching, exception handling, human review, and continuous-learning workflows.
- Architected a **LangGraph Supervisor/Worker multi-agent framework** with confidence-based self-correction and semantic tool routing, improving tool-selection accuracy from **70% to 95%** while reducing token usage by approximately **80%**.
- Designed a reconciliation pipeline combining **structured processing, clustering, deterministic rules, and LLM agents**, achieving **90%+ matching accuracy across 10,000+ daily transactions**.
- Built production **Python/FastAPI AI services**, human-in-the-loop workflows, evaluation, guardrails, failure recovery, and observability, owning delivery from architecture through rollout and operational support.

09/2017 – 01/2020

Remote

- Software Development Engineer II

Amazon

- Modernized **Amazon Payments** by migrating business-critical workloads from **Oracle to DynamoDB**, strengthening the data foundation for scalable, cloud-native transaction processing.
- Redesigned **backend data-access patterns and distributed service boundaries** for high-volume NoSQL workloads, improving scalability, performance, maintainability, and operational flexibility.
- Refactored production services around **modular, testable, and fault-isolated architectures**, improving reliability across business-critical transaction workflows.
- Strengthened **CI/CD, automated testing, deployment reliability, and production operations**, supporting safer delivery and troubleshooting of enterprise-scale distributed systems.

EXPERIENCE

07/2015 – 08/2017

Blackwood, NJ

- Full-Stack & Integration Engineer

Cleo

- Designed and delivered **enterprise B2B integration services** connecting customer platforms with suppliers, trading partners, and major retailers through APIs and structured data workflows.
- Built reusable **EDI, REST, and SOAP services** for document ingestion, transformation, validation, and system-to-system automation across complex enterprise environments.
- Developed full-stack integration tooling for **transaction processing, operational data, external APIs, and workflow automation**, strengthening reliable data exchange across partner systems.
- Supported the **full production lifecycle** from implementation and testing through deployment, troubleshooting, integration support, and operational maintenance.

03/2012 – 12/2014

Philadelphia, PA

- Full-Stack Software Engineer

FreedomPay

- Developed backend and full-stack capabilities for an **enterprise payments platform**, including transaction workflows, APIs, data integrations, and merchant-facing applications.
- Built reusable **API and integration services** connecting payment systems, POS platforms, merchant applications, and third-party services.
- Supported **high-volume transaction-processing workflows** with strong requirements around data integrity, availability, performance, security, and production reliability.
- Worked across implementation, integration, testing, deployment, and operational support, building a strong foundation in **production backend and distributed-system engineering**.

10/2011 – 01/2012

Philadelphia, PA

- Software Developer & Scrum Master

Elsevier

- Contributed directly to **application development, debugging, testing, and technical problem solving** while coordinating delivery across engineering and product stakeholders.
- Translated business requirements into **clear user stories, technical priorities, and executable development plans**, keeping implementation aligned with product needs.
- Facilitated **sprint planning, backlog refinement, standups, retrospectives, and release coordination** while remaining actively involved in software delivery.
- Helped strengthen **release readiness, software quality, team communication, and delivery predictability** through disciplined Agile engineering practices.

EDUCATION

2007 – 2009

- Master's Degree, Computer Science

Princeton University

2003 – 2007

- Bachelor's Degree, Computer Science

Stevens Institute of Technology

TECH STACK

Languages: Python, TypeScript, JavaScript, SQL, Java, PHP
AI / GenAI: LangGraph, LLMs, RAG, Agentic AI, Multi-Agent Systems, OpenAI, Gemini, Claude, Prompt Engineering
Python / Backend: FastAPI, REST APIs, Microservices, Async Processing, API Integration, Structured Outputs
Data / Cloud: PostgreSQL, DynamoDB, Redis, Kafka, SQL/NoSQL, Vector Search, Embeddings, AWS, Serverless
DevOps / Frontend: Kubernetes, Docker, CI/CD, Observability, Automated Testing, React, Next.js, TypeScript